

# FREQFORMER: EFFICIENT POLYP SEGMENTATION VIA WAVELET TRANSFORM

Xudong Zhou<sup>1,2</sup>, Tianxiang Chen<sup>2\*</sup>

<sup>1</sup> Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei 230031, China

<sup>2</sup> University of Science and Technology of China, Hefei 230026, China

## ABSTRACT

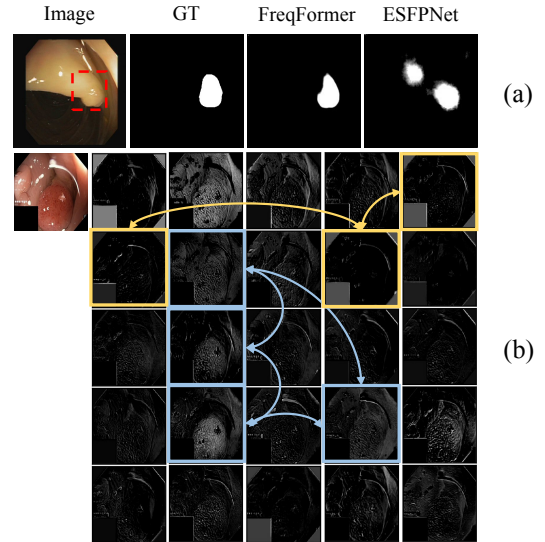
Polyp segmentation is crucial for detecting colorectal cancer. Despite some progresses, present methods struggle with shape-aware segmentation due to minimal gradient discrepancies between the polyp and background, causing blurry edges and lack efficiency due to feature map redundancy. We analyze the reason for the first challenge is that most methods either only rely on spatial domain features while ignoring frequency domain clues, or ignore interactions between the two domains. Frequency clues are sensitive to the minor gradient discrepancies in spatial domain, which helps shape-aware segmentation. Based on this, we propose Efficient Frequency-Space Interaction Attention (EFSIA) to interact spatial features with high and low frequency clues and further strengthen the interaction in the decoder by devising Fourier Cross Fusion Module (FCFM). To tackle the second challenge, we reduce the feature map redundancy with proposed EFSIA by removing similar feature maps and using efficient wavelet transform to recover them without accuracy drop. Experiments on Kvasir-SEG and CVC-ClinicDB prove the superiority of our method in accuracy and efficiency.

**Index Terms**— Polyp segmentation, Wavelet domain, Computational efficiency

## 1. INTRODUCTION

Polyp segmentation aims to segment polyps from endoscopic images of the colon and is a crucial step in the early detection of colorectal cancer. Therefore, how to achieve accurate and efficient polyp segmentation really matters.

Despite some recent progresses [1, 2], present methods lack efficiency since many of their feature maps are redundant. This is visualized in Fig. 1 (b) where numerous very similar feature maps exist. Notably, the process of generating all these feature maps is also computationally expensive due to multiple convolution operations. Furthermore, present methods still fail to achieve accurate shape-aware segmentation since the irregular polyps always seem so similar with surrounding tissues that their boundaries are blurry. Given that frequency features are more sensitive to the minor gradient discrepancies than spatial ones, we argue that the main

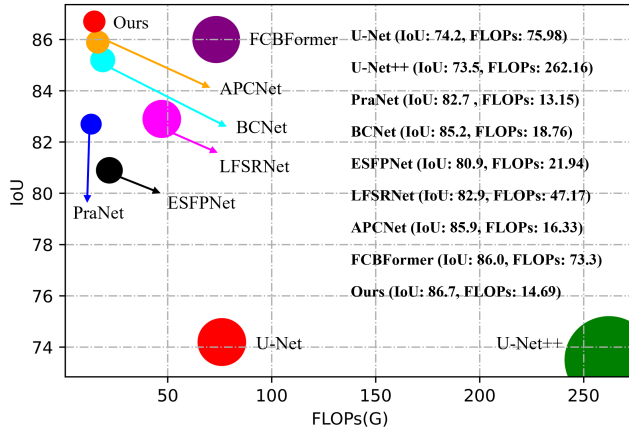


**Fig. 1.** Visualization of (a) polyp-background similarity and (b) feature map redundancy challenges. The feature maps in (b) are generated by [3], where similar feature map examples are annotated with boxes in the same color.

reason is that present polyp segmentation methods only consider spatial features while ignoring frequency clues or ignoring sufficient interaction between the two domains. Our method efficiently interacts frequency clues of different bands with spatial features. By comparing our method with a recent SOTA method [2] that only uses spatial features, as shown in Fig. 1 (a), we can observe that the polyps hard to completely segment in spatial domain can be finely segmented by our method in a shape-aware manner with the help of frequency clues. This observation complies with our argument.

To tackle the above two challenges, we propose FreqFormer for efficient shape-aware polyp segmentation. To distinguish polyp pixels from similar background, we devise Efficient Frequency-Space Interaction Attention (EFSIA) and embed it to Efficient Frequency-Space Interaction Block (EFSIB) to partition the whole image frequencies to high frequency edge details and low frequency basic structures. In this way, the minor gradient discrepancies in RGB spatial domain between polyp and similar background can be ampli-

\*Tianxiang Chen is the corresponding author.



**Fig. 2.** Performance vs computational efficiency on KVASIR-SEG. Our method achieves the best trade-off between efficiency and performance. Larger dots represent higher FLOPs.

fied and preserved to different frequency intervals. Moreover, for efficiency we filter some similar feature maps and employ the efficient wavelet transform to obtain high and low frequency band features and interact with the spatial features to recover the filtered redundant information without performance drop. To further boost cross-layer frequency-space feature interaction, at the junction of encoder and decoder we design a plug and play Fourier Cross Fusion Module (FCFM), where Fast Fourier Convolution (FFConv) is introduced so that frequency and spatial domain features can be simultaneously extracted. Fig. 2 shows performance vs computational efficiency on KVASIR-SEG.

Our main contributions are summarized as follows:

- We interact spatial and frequency features in a cost-effective approach to realize shape-aware polyp segmentation and reduce feature redundancy.
- We propose Efficient Frequency-Space Interaction Attention (EFSIA) in the encoder to efficiently distinguish polyps from similar background by interacting spatial features with the high and low frequency clues divided by efficient wavelet transform. We also propose Fourier Cross Fusion Module (FCFM) for further space-frequency interactions in the decoder.
- Experiments on two datasets show that our method achieves SOTA results with low computational cost.

## 2. RELATED WORK

### 2.1. Polyp Segmentation

The past years have witnessed the thrive of CNNs in segmentation tasks because of their powerful feature capture ability.

Among them, U-shaped structures [3, 4, 5, 6] take the lead. However, the limited receptive field of CNN makes it hard to capture long-range dependencies, so the segmentation is often incomplete when the polyp covers large image areas. Since ViT [7] excels at modelling long-range dependencies, some ViT-based methods [8, 9] that combine ViT with CNN have shown stronger performances. Despite these progresses, present methods still face challenges in effectively addressing the issue of polyp-background similarity, resulting in false alarms or incomplete results. Moreover, these methods often come with high computational costs [10], which further adds to the complexity and limitations of the existing approaches. To address this, we propose FreqFormer, which incorporates the Efficient Frequency-Space Interaction Attention (EFSIA) and Fourier Cross Fusion Module (FCFM).

### 2.2. Frequency Based Representation Learning

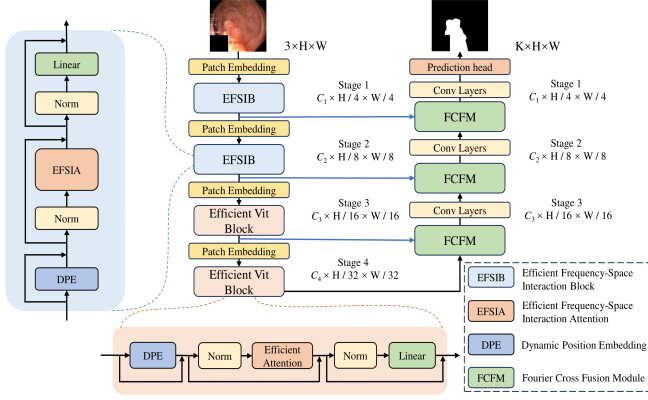
Frequency based representation learning has been involved in both natural image and medical image segmentation tasks. HiLo [11] has achieved good performance in natural image by frequency-based transformer structure. However, it merely applies basic multi head self-attention and average pooling to input feature map respectively to get high frequencies and low frequencies, so its high-frequency branch is only "high" relative to its low frequency branch but actually is a spatial attention branch. Also, its down-sampling operation in attention will inevitably cause information loss. Wave-ViT [12] solves this by applying DWT and inverse DWT but does not make a distinction between high and low frequency features. P-Mamba [13] devises a DWT-based module that can de-noise while preserving target boundaries.

Dividing Frequency into different bands can better help shape-aware polyp segmentation, as high frequencies reflect detailed edge and texture information of the object, while low frequencies capture the overall structure of the object. As demonstrated by Reza Azad et al. [14], they utilized methods such as Laplacian pyramids to partition the frequency domain and achieved promising performance on the skin disease dataset. However, they rarely care about network efficiency, though their multi-stage features extracted by multiple convolutions or attentions inevitably incurs extra feature redundancy and computational costs. Moreover, They also neglect interacting with spatial domain features while handling frequency domain information. Different from above methods, we tackle these issues by proposing the Efficient Frequency-Space Interaction Attention (EFSIA).

## 3. METHOD

### 3.1. Overview

Our FreqFormer follows a U-shaped encoder-decoder structure, within which we devise two novel modules to help segmenting polyps via wavelet domain and spatial domain. In

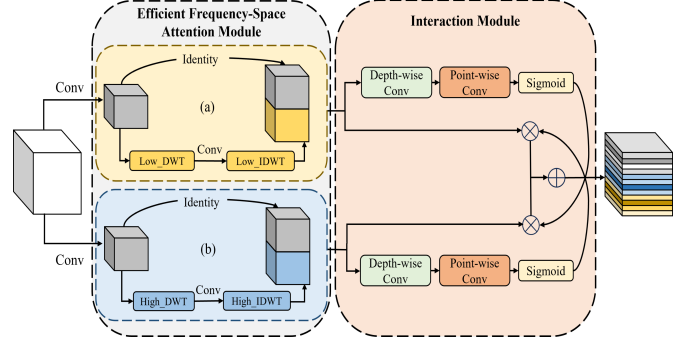


**Fig. 3.** Overview of our proposed FreqFormer for efficient shape-aware polyp segmentation.

the encoder, we introduce Efficient Frequency-Space Interaction Attention (EFSIA). This lightweight attention module is employed for feature extraction of both spatial and wavelet domains, and it is embedded within the Efficient Frequency-Space Interaction Block (EFSIB). We set the number of EFSIB to 3 in stage 1 and 4 in stage 2. These blocks are primarily focused on capturing the local features of the image. Stage 3 and Stage 4 each consist of 8 and 3 Efficient ViT Block, which are responsible for extracting global features. In order to alleviate the issues of feature redundancy and computational complexity associated with self-attention, we embed the Efficient Attention [15] within the Efficient ViT Block. This attention mechanism can provide an equivalent representation of self-attention while being computationally efficient. At the junction of encoder and decoder of each stage, we devise Fourier Cross Fusion Module (FCFM) to realize cross layer feature fusion in spatial and frequency domain for further refinement. The decoder progressively enlarges the fused features by four stages until reaching a segmentation head to generate the final mask result. Fig. 3 shows overview of our proposed FreqFormer.

### 3.2. Efficient Frequency-Space Interaction Attention

In the wavelet domain of the image, the low-frequency component usually reflects the basic structure of the object at coarse-grained level, while the high-frequency component presents the edge information of the object at fine-grained level. In order to segment polyp pixels from similar background. Inspired by the Ghost module [16], we incorporate Discrete Wavelet Transform (DWT) into the convolution design. This allows us to analyze the image at different frequency scales. Additionally, we combine spatial domain information to propose the Efficient Frequency-Space Interaction Attention, which is shown in Fig. 4. The intuition is that partitioning frequency information from the wavelet domain helps to highlight the minor gradient discrepancies in



**Fig. 4.** Structure of Efficient Frequency-Space Interaction Attention (EFSIA). (a) Efficient Low Frequency-Space Combination (ELFSC). (b) Efficient High Frequency-Space Combination (EHFSC).

images. With DWT, we can partition the input feature to high frequency branch preserving edge details and low frequency branch preserving overall structure information. The deep interaction of spatial and frequency component contributes to better shape-aware polyp segmentation.

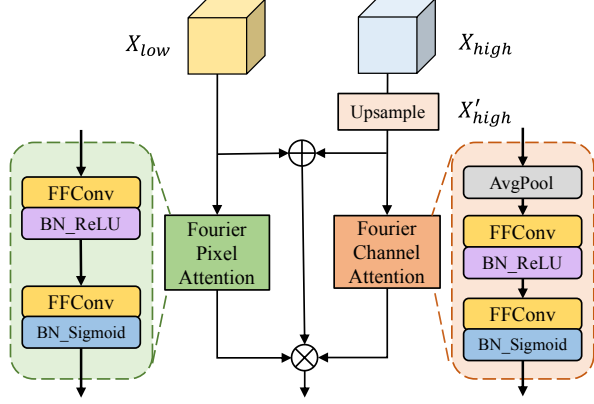
Given an input feature map  $I \in R^{H \times W \times C}$ , we first transform it into  $I_1$  and  $I_2$  by convolution operation, where  $I_1$  and  $I_2 \in R^{H \times W \times C'}$ ,  $C' \leq C$ . First, we use the Haar Discrete Wavelet Transform to separately decompose  $I_1$  and  $I_2$  into low-frequency component  $I'_L$  and high-frequency component  $I'_H$ . Specifically, DWT first utilizes a high-pass filter  $f_H = (1/\sqrt{2}, -1/\sqrt{2})$  and a low-pass filter  $f_L = (1/\sqrt{2}, 1/\sqrt{2})$  to encode feature map into a high-frequency sub-band  $I_H$  and a low-frequency sub-band  $I_L$ , respectively. Next,  $f_L, f_H$  are applied again to decompose  $I_L$  and  $I_H$  into four wavelet sub-bands:  $I_{LH}, I_{HL}, I_{HH}$  and  $I_{LL}$ , all with  $R^{H/2 \times W/2 \times C'/4}$  in size. To get high-frequency component  $I'_H$ , we concatenate  $I_{LH}, I_{HL}, I_{HH}$  along the channel dimension. The remained  $I_{LL}$  represents the low frequency component  $I'_L$ . And then, convolution operation is applied to extract corresponding frequency band features. Finally, we transform the low-frequency component and high-frequency component back to the spatial domain using the Inverse Discrete Wavelet Transform (IDWT). We then concatenate them with  $I_1$  and  $I_2$  respectively. In this way, we can get the partitioned frequency-space feature map aggregated in the spatial domain. The Efficient Frequency-Space Attention Module is computed as follows:

$$ELFSC = \text{Concat} [I_1, \phi_1 (I_1)] \quad (1)$$

$$EHFSC = \text{Concat} [I_2, \phi_2 (I_2)] \quad (2)$$

where  $\phi_1$  denotes the operations of retaining only the low-frequency components by DWT, convolution, and IDWT.  $\phi_2$  denotes the operations of retaining only the high-frequency components by DWT, convolution, and IDWT.

Then, by the Interaction Module, we perform cross interaction of low frequency-space and high frequency-space, re-



**Fig. 5.** Structure of Fourier Cross Fusion Module (FCFM).

sulting in a feature map that simultaneously possesses refined spatial and frequency information. The specific calculation method of the Interaction Module is as follows:

$$\mathbf{IM}(I_l, I_h) = \sigma(DSC(I_l)) * I_h + \sigma(DSC(I_h)) * I_l \quad (3)$$

where  $I_l$  and  $I_h$  represent the outputs of ELFSC and EHFSC, respectively.  $DSC$  represents depthwise separable convolution,  $\sigma$  represents *sigmoid* activation function, and  $*$  represents Hadamard product.

### 3.3. Fourier Cross Fusion Module

We devise Fourier Cross Fusion Module (FCFM) based on the Fast Fourier Convolution [17] to further integrate spatial contextualized information and frequency domain contextualized information of hierarchical multi-scale feature maps. The structure of the FCFM is shown in Fig. 5.

The inputs of FCFM are the low-level feature  $X_{low}$  from the encoder and the high-level feature  $X_{high}$  from the decoder.  $X_{high}$  is first up-sampled via transposed convolution to the same size as  $X_{low}$  and we denote the up-sampled  $X_{high}$  as  $X'_{high}$ .  $X_{low}$  contains polyp location and detail information so we feed it into Fourier Pixel Attention ( $\mathbf{A}_{fp}$ ).  $X'_{high}$  is fed into Fourier Channel Attention ( $\mathbf{A}_{fc}$ ) to select the most important channels. After that, we add  $X'_{high}$  and  $X_{low}$ .  $\mathbf{A}_{fp}$  and  $\mathbf{A}_{fc}$  are constrained separately after the features are directly fused by sum, as shown in Equation (4). Finally, We Hadamard product the output of the three parts as the output of FCFM, which is formulated as:

$$\mathbf{FCFM} = \mathbf{A}_{fc}(X_{high}) * \mathbf{A}_{fp}(X_{low}) * (X'_{high} + X_{low}), \quad (4)$$

### 3.4. Loss Function

We use Binary cross-entropy loss and Dice loss for supervised training, the overall loss is computed as follows:

$$\mathcal{L}_{bce} = - \sum_{i \in I} y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \quad (5)$$

$$\mathcal{L}_{dice} = 1 - \frac{2 \sum_{i \in I} y_i \hat{y}_i}{\sum_{i \in I} y_i + \sum_{i \in I} \hat{y}_i} \quad (6)$$

$$\mathcal{L}_{loss} = \mathcal{L}_{bce} + \lambda \mathcal{L}_{dice} \quad (7)$$

where  $y$  and  $\hat{y}_i$  represent the ground truth and output, respectively.  $i \in I$  refers to a pixel in the  $y$  and  $\hat{y}_i$ , and the default setting of  $\lambda$  is 1.

## 4. EXPERIMENT

### 4.1. Datasets

**Kvasir-SEG dataset** [18]. This is a largest-scale challenging dataset for gastrointestinal polyp segmentation containing 1000 polyp images with sizes varying from  $332 \times 487$  to  $1920 \times 1072$ . Since the official split setting for training and testing is not provided, we follow the same split setting as in [19]: 80% for training and 20% for testing.

**CVC-ClinicDB dataset** [20]. CVC-clinicDB contains 612 polyp images from 29 different colonoscopy sequences with  $384 \times 288$  in size. Following the split of Kvasir-SEG, we randomly select 490 images for training and 122 for testing.

### 4.2. Evaluation Metrics

For the systematical performance evaluation metrics of the model. We use MAE, Dice and IoU, where MAE means Mean Absolute Error, Dice and IoU represent Dice Similarity Coefficient and Intersection-over-Union respectively. Among these metrics, Dice and IoU are used to measure the similarity between the predicted mask and the ground truth. Formally, they can be computed as follows:

$$Dice = \frac{2TP}{(TP + FN) + (TP + FP)} \quad (8)$$

$$IoU = \frac{TP}{FN + FP + TP} \quad (9)$$

where FN, TP, and FP represent false negative, true positive and false positive predictions, respectively.

### 4.3. Experimental Settings

We use vertical flips, random rescales, cutout and horizontal flips as data augmentations to improve model generalization ability. All experiments were implemented in the Pytorch and performed on NVIDIA GeForce RTX 3090 GPU. The model was iterated for 300 epochs using the AdamW optimizer with



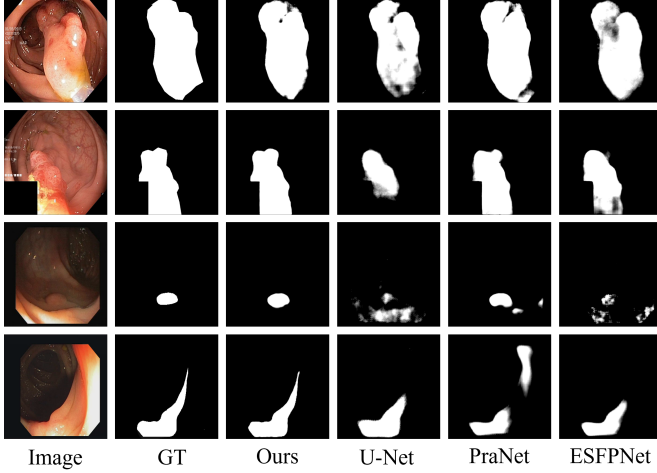


Fig. 6. Visual comparison with different methods.

initial learning rate  $10^{-4}$ , weight decay  $10^{-4}$ , which is reduced by the “Poly” strategy. The batch size for training is 8. All images and masks are resized to  $512 \times 512$  during training and testing.

## 5. RESULT

### 5.1. Comparison With The State-of-the-arts

As shown in Table 1, our FreqFormer achieves the best performance in terms of Dice and IoU compared with other most recent methods. Some visual results of polyp-background similarity cases are given in Fig. 6, where we can observe that FreqFormer precisely delineate polyp shapes from similar background while other methods fails to do so.

Table 1. Comparison with the SOTA methods.

| Method            | Kvasir-SEG      |                |                  | CVC-ClinicDB    |                |                  |
|-------------------|-----------------|----------------|------------------|-----------------|----------------|------------------|
|                   | Dice $\uparrow$ | IoU $\uparrow$ | MAE $\downarrow$ | Dice $\uparrow$ | IoU $\uparrow$ | MAE $\downarrow$ |
| U-Net             | 82.8            | 74.2           | 0.050            | 87.0            | 79.6           | 0.018            |
| U-Net++           | 81.8            | 73.5           | 0.053            | 81.7            | 75.3           | 0.024            |
| PraNet            | 88.6            | 82.7           | 0.031            | 90.3            | 85.1           | 0.011            |
| DoubleU-Net       | 80.1            | 72.1           | 0.064            | 74.4            | 66.2           | 0.041            |
| BCNet             | 90.5            | 85.2           | 0.029            | 91.1            | 86.3           | 0.008            |
| ESFPNet           | 87.8            | 80.9           | 0.040            | 91.0            | 85.8           | 0.011            |
| LFSRNet           | 88.3            | 82.9           | 0.030            | 90.2            | 85.5           | 0.009            |
| APCNet            | 91.3            | 85.9           | 0.023            | 93.4            | 88.6           | <b>0.008</b>     |
| FCBFormer         | 91.4            | 86.0           | 0.024            | 89.1            | 84.0           | 0.015            |
| FreqFormer (Ours) | <b>91.8</b>     | <b>86.7</b>    | <b>0.023</b>     | <b>93.7</b>     | <b>89.7</b>    | 0.009            |

### 5.2. Spectrum Analysis of FreqFormer

Fig. 7 shows the magnitude ( $14 \times 14$ ) of the frequency component [11] for mean of 100 samples by applying Fast Fourier Transform (FFT) to the 5 output channels of the Efficient Low Frequency-Space Combination (ELFSC) and Efficient

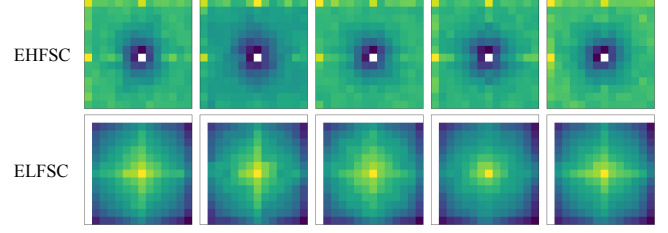


Fig. 7. Visualization of frequency magnitude.

High Frequency-Space Combination (EHFSC). In the visualization, lighter colors represent larger magnitudes and pixels closer to the center indicate lower frequencies. The visualization results indicate that EHFSC primarily captures high frequencies while ignoring low frequencies, and ELFSC mainly focuses on low frequencies. This demonstrates a clear disentanglement of high and low frequencies within a single attention layer.

### 5.3. Computational Efficiency Comparison

We also compare FreqFormer with recent methods from the aspect of computational efficiency. The comparison results in terms of floating-point operations (FLOPs) and the number of parameters (Params) are presented in Table 2. Among these methods, PraNet requires the fewest FLOPs, while U-Net has the fewest Params. However, they achieve mediocre segmentation performance and are outperformed by our FreqFormer in quantitative comparisons, as demonstrated in Tables 1. Our method achieves a balance between accuracy and model complexity. It is worth noting that this impressive characteristic makes it highly adaptable for both accuracy-oriented and efficiency-oriented applications.

Table 2. Computational efficiency comparison.

| Method      | Params $\downarrow$ | FLOPs $\downarrow$ | Methods           | Params $\downarrow$ | FLOPs $\downarrow$ |
|-------------|---------------------|--------------------|-------------------|---------------------|--------------------|
| U-Net       | <b>17.27M</b>       | 75.98G             | ESFPNet           | 61.69M              | 21.94G             |
| U-Net++     | 36.63M              | 262.16G            | LFSRNet           | 48.39M              | 47.17G             |
| PraNet      | 30.50M              | <b>13.15G</b>      | APCNet            | 33.11M              | 16.33G             |
| DoubleU-Net | 144.06M             | 2.459T             | FCBFormer         | 52.94M              | 73.30G             |
| BCNet       | 26.82M              | 18.76G             | FreqFormer (Ours) | 26.56M              | 14.69G             |

## 6. CONCLUSION

We propose FreqFormer for polyp segmentation to polyp-background similarity challenge via wavelet transform. Within FreqFormer we devise Efficient Frequency-Space Interaction Attention (EFSIA) to highlight blurry polyp boundaries from similarity and Fourier Cross Fusion Module (FCFM) for feature fusion. Experiments on two datasets show that our method outperforms others with low computational cost.

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